

Solutions – Midterm Exam 1

ECON 320 – Introduction to Mathematical Economics

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Instructor: Muhebullah Karimzada

1. Let $U = \mathbb{R}$ and

$$A = \{x \in \mathbb{R} : x^2 - 7x + 12 \geq 0\}, \quad B = \{x \in \mathbb{R} : |x - 4| \leq 3\}, \quad C = \{x \in \mathbb{R} : 1 < x < 5\}.$$

(a) Write A , B , C as unions of intervals.

Factor $x^2 - 7x + 12 = (x - 3)(x - 4)$.

$$(x - 3)(x - 4) \geq 0 \iff x \leq 3 \text{ or } x \geq 4.$$

Hence $A = (-\infty, 3] \cup [4, \infty)$.

For B : $|x - 4| \leq 3 \iff -3 \leq x - 4 \leq 3 \iff 1 \leq x \leq 7$, so $B = [1, 7]$.

For C : $C = (1, 5)$.

(b) Compute $A \cap B$, $A \cup C$, $B \setminus C$.

$$A \cap B = ((-\infty, 3] \cup [4, \infty)) \cap [1, 7] = [1, 3] \cup [4, 7].$$

$$A \cup C = ((-\infty, 3] \cup [4, \infty)) \cup (1, 5) = (-\infty, 3] \cup (1, 5) \cup [4, \infty) = \mathbb{R}.$$

$$B \setminus C = [1, 7] \setminus (1, 5) = \{1\} \cup [5, 7] = [1, 1] \cup [5, 7].$$

(c) Verify $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$.

This is a standard identity: for any sets X, Y, Z ,

$$(X \cup Y) \setminus Z = (X \setminus Z) \cup (Y \setminus Z).$$

Proof: If $x \in (X \cup Y) \setminus Z$, then $(x \in X \text{ or } x \in Y)$ and $x \notin Z$, so $x \in (X \setminus Z) \cup (Y \setminus Z)$. Conversely, if $x \in (X \setminus Z) \cup (Y \setminus Z)$, then either $x \in X$ or $x \in Y$ and $x \notin Z$, hence $x \in (X \cup Y) \setminus Z$. Thus equality holds.

2. Let

$$u(x, y) = x^2y - y^3 + \ln(1 + xy), \quad x = r^2, \quad y = s^3, \quad r > 0, \quad s > 0.$$

(a) Compute $\partial u / \partial x$ and $\partial u / \partial y$.

Differentiate term by term:

$$u_x = 2xy + \frac{y}{1 + xy}, \quad u_y = x^2 - 3y^2 + \frac{x}{1 + xy}.$$

(b) Compute $\partial u / \partial r$ and $\partial u / \partial s$ in terms of r, s .

Chain rule:

$$\begin{aligned} \frac{\partial u}{\partial r} &= u_x \frac{\partial x}{\partial r} + u_y \frac{\partial y}{\partial r} = u_x \cdot (2r) + u_y \cdot 0 = 2r u_x \Big|_{(x,y)=(r^2,s^3)} \\ \Rightarrow \quad \frac{\partial u}{\partial r} &= 2r \left(2(r^2)s^3 + \frac{s^3}{1 + r^2s^3} \right) = 4r^3s^3 + \frac{2rs^3}{1 + r^2s^3}. \end{aligned}$$

Similarly,

$$\begin{aligned} \frac{\partial u}{\partial s} &= u_x \cdot 0 + u_y \cdot \frac{\partial y}{\partial s} = u_y \cdot (3s^2) = 3s^2 u_y \Big|_{(x,y)=(r^2,s^3)} \\ \Rightarrow \quad \frac{\partial u}{\partial s} &= 3s^2 \left(r^4 - 3s^6 + \frac{r^2}{1 + r^2s^3} \right). \end{aligned}$$

3. Let $F(x, y) = \ln(y + 2x) - x^2 + 3y - 10$ and suppose $F(x, y) = 0$ defines y as a function of x near $x = 1$.

(a) Find y such that $F(1, y) = 0$ and verify conditions.

At $x = 1$,

$$F(1, y) = \ln(y + 2) - 1 + 3y - 10 = \ln(y + 2) + 3y - 11.$$

We need $\ln(y + 2) + 3y - 11 = 0$ with $y > -2$ for domain. By inspection: $y = 3$ gives $\ln 5 + 9 - 11 \approx 1.609 - 2 = -0.391$; $y = 3.12$ gives $\ln(5.12) + 9.36 - 11 \approx 1.634 + (-1.64) = -0.006$; $y \approx 3.121$ gives ≈ 0 . So the root is $y^* \approx 3.121$.

Implicit function theorem requires $F_y(1, y^*) \neq 0$. Compute

$$F_x = \frac{2}{y + 2x} - 2x, \quad F_y = \frac{1}{y + 2x} + 3.$$

At $(1, y^*)$, $F_y = \frac{1}{y^*+2} + 3 > 0$, hence an implicit function exists locally.

(b) Compute $\frac{dy}{dx}$ at $(1, y^*)$.

By implicit differentiation,

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{\frac{2}{y+2x} - 2x}{\frac{1}{y+2x} + 3}.$$

At $(1, y^*)$ with $y^* \approx 3.121$, we have $y^* + 2 \approx 5.121$,

$$F_x \approx \frac{2}{5.121} - 2 \approx 0.3906 - 2 = -1.6094, \quad F_y \approx \frac{1}{5.121} + 3 \approx 0.1953 + 3 = 3.1953.$$

Thus

$$\left. \frac{dy}{dx} \right|_{(1, y^*)} \approx -\frac{-1.6094}{3.1953} \approx 0.504.$$

(c) Linear approximation for $x : 1 \rightarrow 1.04$.

Total differential $dF = F_x dx + F_y dy = 0$ gives

$$dy \approx -\frac{F_x}{F_y} dx \approx 0.504 \times 0.04 \approx 0.0202.$$

So y increases by about 0.020 when x rises from 1 to 1.04.

4. Transformation $(u, v, w) \mapsto (x, y, z)$:

$$x = u^2 + 2vw, \quad y = v^2 + 2uw, \quad z = w^2 + 2uv.$$

(a) Compute $J = \partial(x, y, z)/\partial(u, v, w)$.

Partial derivatives:

$$\begin{aligned} \frac{\partial x}{\partial u} = 2u, \quad \frac{\partial x}{\partial v} = 2w, \quad \frac{\partial x}{\partial w} = 2v; \quad \frac{\partial y}{\partial u} = 2w, \quad \frac{\partial y}{\partial v} = 2v, \quad \frac{\partial y}{\partial w} = 2u; \\ \frac{\partial z}{\partial u} = 2v, \quad \frac{\partial z}{\partial v} = 2u, \quad \frac{\partial z}{\partial w} = 2w. \end{aligned}$$

Hence

$$J = \begin{bmatrix} 2u & 2w & 2v \\ 2w & 2v & 2u \\ 2v & 2u & 2w \end{bmatrix} = 8 \cdot \frac{1}{8} \begin{bmatrix} 2u & 2w & 2v \\ 2w & 2v & 2u \\ 2v & 2u & 2w \end{bmatrix}.$$

It is convenient to factor 2 from each row:

$$\det(J) = 2^3 \det \begin{bmatrix} u & w & v \\ w & v & u \\ v & u & w \end{bmatrix}.$$

Call the 3×3 matrix above M .

Compute $\det(M)$ by expansion along the first row:

$$\det(M) = u \begin{vmatrix} v & u \\ u & w \end{vmatrix} - w \begin{vmatrix} w & u \\ v & w \end{vmatrix} + v \begin{vmatrix} w & v \\ v & u \end{vmatrix}.$$

Evaluate minors:

$$\begin{vmatrix} v & u \\ u & w \end{vmatrix} = vw - u^2, \quad \begin{vmatrix} w & u \\ v & w \end{vmatrix} = w^2 - uv, \quad \begin{vmatrix} w & v \\ v & u \end{vmatrix} = uw - v^2.$$

Thus

$$\det(M) = u(vw - u^2) - w(w^2 - uv) + v(uw - v^2) = 3uvw - (u^3 + v^3 + w^3).$$

Therefore

$$\boxed{\det(J) = 8(3uvw - u^3 - v^3 - w^3)}.$$

(b) Evaluate at $(u, v, w) = (1, -2, 3)$ and test invertibility.

Plugging in:

$$3uvw = 3 \cdot 1 \cdot (-2) \cdot 3 = -18, \quad u^3 + v^3 + w^3 = 1 + (-8) + 27 = 20.$$

Hence

$$\det(J) = 8(-18 - 20) = 8(-38) = -304 \neq 0.$$

Conclusion: J is nonsingular, so the mapping is locally invertible at $(1, -2, 3)$ (inverse function theorem).

5. Utility

$$U(x_1, x_2, \ell) = \ln x_1 + \ln x_2 + \beta \ln \ell, \quad \beta > 0,$$

budget/time constraint

$$p_1 x_1 + p_2 x_2 + w \ell = I,$$

and first-order conditions (given)

$$\begin{cases} F_1 = p_1 x_1 + p_2 x_2 + w \ell - I = 0, \\ F_2 = \frac{1}{x_1} - \lambda p_1 = 0, \\ F_3 = \frac{1}{x_2} - \lambda p_2 = 0. \end{cases}$$

We treat (x_1, x_2, ℓ) as endogenous for comparative statics, with parameters (p_1, p_2, w, I) and λ entering F_2, F_3 as given.

(a) Jacobian $J = \partial(F_1, F_2, F_3)/\partial(x_1, x_2, \ell)$.

$$\frac{\partial F_1}{\partial(x_1, x_2, \ell)} = (p_1, p_2, w), \quad \frac{\partial F_2}{\partial(x_1, x_2, \ell)} = \left(-\frac{1}{x_1^2}, 0, 0\right), \quad \frac{\partial F_3}{\partial(x_1, x_2, \ell)} = \left(0, -\frac{1}{x_2^2}, 0\right).$$

Thus

$$J = \begin{bmatrix} p_1 & p_2 & w \\ -\frac{1}{x_1^2} & 0 & 0 \\ 0 & -\frac{1}{x_2^2} & 0 \end{bmatrix}.$$

(b) Evaluate J and $\det(J)$ at $(p_1, p_2, w, \beta, I) = (2, 1, 1, 1, 12)$ and $(x_1, x_2, \ell) = (2, 4, 6)$.

Substitute $p_1 = 2, p_2 = 1, w = 1, x_1 = 2, x_2 = 4$:

$$J = \begin{bmatrix} 2 & 1 & 1 \\ -\frac{1}{4} & 0 & 0 \\ 0 & -\frac{1}{16} & 0 \end{bmatrix}.$$

Compute determinant (expand along third column; only the (1, 3) entry is nonzero):

$$\det(J) = 1 \cdot \det \begin{bmatrix} -\frac{1}{4} & 0 \\ 0 & -\frac{1}{16} \end{bmatrix} = \left(-\frac{1}{4}\right) \left(-\frac{1}{16}\right) - 0 = \frac{1}{64} > 0.$$

J is nonsingular at the given bundle.

(c) Signs of $\frac{\partial x_1}{\partial I}, \frac{\partial x_2}{\partial I}, \frac{\partial \ell}{\partial I}$ via Cramer/sign test.

Treat I as the parameter. The derivative system comes from differentiating $F(\cdot) = 0$ w.r.t. I :

$$J \begin{bmatrix} \frac{\partial x_1}{\partial I} \\ \frac{\partial x_2}{\partial I} \\ \frac{\partial \ell}{\partial I} \end{bmatrix} = - \begin{bmatrix} \frac{\partial F_1}{\partial I} \\ \frac{\partial F_2}{\partial I} \\ \frac{\partial F_3}{\partial I} \end{bmatrix} = - \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Let $v = (x_{1I}, x_{2I}, \ell_I)^\top$. Solve $Jv = (1, 0, 0)^\top$:

From row 2: $-(1/x_1^2) x_{1I} = 0 \Rightarrow x_{1I} = 0$.

From row 3: $-(1/x_2^2) x_{2I} = 0 \Rightarrow x_{2I} = 0$.

From row 1: $p_1 x_{1I} + p_2 x_{2I} + w \ell_I = 1 \Rightarrow w \ell_I = 1 \Rightarrow \ell_I = 1/w > 0$.

Therefore,

$$\begin{bmatrix} \frac{\partial x_1}{\partial I} = 0, & \frac{\partial x_2}{\partial I} = 0, & \frac{\partial \ell}{\partial I} = \frac{1}{w} > 0. \end{bmatrix}$$

Interpretation. Given the three-equation system specified (budget and two FOCs for x_1, x_2), a marginal increase in income is absorbed entirely by leisure to maintain the FOCs $1/x_i = \lambda p_i$ with the same λ entering F_2, F_3 ; the goods do not adjust at the margin in this restricted system. With a full system including the leisure FOC $U_\ell - \lambda w = 0$, all three would typically increase with I in Cobb–Douglas.